

15 — Features Implemented & System Capabilities Checklist

■ Comprehensive Feature Implementation Status

This document provides a feature checklist evaluating every major component against codebase evidence.

■ Feature Implementation Matrix

Feature / Capability	Implementation Status	Evidence in Codebase (<i>CODE_PH_0</i>)	Operational Summary
■ Predictive Cargo Volume Forecasting	Implemented	<i>CODE_PH_0</i>	Dual Holt-Winters + Ridge ML time-series model projecting 3, 6, 12-month cargo tonnage with 95% dynamic confidence bounds.
Commodity-Wise Traffic Analysis	Implemented	<i>CODE_PH_0</i> , <i>CODE_PH_1</i>	Filters and analyzes 16+ commodity categories (Crude, Coal, Containers, Fertilizers, POL, LPG, etc.).
Import / Export Directional Filtering	Implemented	<i>CODE_PH_0</i> , <i>CODE_PH_1</i>	Filters traffic by <i>CODE_PH_0</i> (Export) vs <i>CODE_PH_1</i> (Import) vs <i>CODE_PH_2</i> .
Model Validation & Backtesting	Implemented	<i>CODE_PH_0</i> , <i>CODE_PH_1</i>	Chronological out-of-sample backtesting evaluating WAPE, MAPE, MAE, RMSE vs Seasonal Naive baseline.
Explainable Forecast Drivers	Implemented	<i>CODE_PH_0</i> , <i>CODE_PH_1</i>	Decomposes forecasts into percentage weights for monsoon, hinterland, and market factors.
Interactive What-If Cargo Simulator	Implemented	<i>CODE_PH_0</i> , <i>CODE_PH_1</i>	Simulates throughput delta, net volume delta, and capacity stress levels from vessel/weather/demand shifts.
Commodity Demand by Trade Lane	Implemented	<i>CODE_PH_0</i> , <i>CODE_PH_1</i>	Analyzes trade route momentum, YoY growth rates, and commodity price indices.

Cargo Anomaly & Risk Intelligence	Implemented	<i>CODE_PH_0, CODE_PH_1</i>	Multi-algorithm anomaly classification (Surge, Decline, Disruption) with 4-tier severity indicator.
NMPA Terminal & Draft Depth Mapping	Implemented	<i>CODE_PH_0</i>	Maps commodities to NMPA berths (OJ-1/OJ-2, Berths 15 & 16, Berth 14 Container) and draft depths.
Multi-Year Historical Datasets	Implemented	<i>CODE_PH_0</i> (1,354 rows)	5-year monthly cargo traffic spanning 2021, 2023, 2024, 2025, and 2026.
Custom CSV File Upload Endpoint	Implemented	<i>CODE_PH_0</i>	Multipart form POST handler for uploading new CSV data files to <i>CODE_PH_0</i> directory.
Interactive Plotly Visualizations	Implemented	<i>CODE_PH_0</i> (Multiple Plotly render blocks)	Time-series lines, confidence shaded bands, horizontal bar charts, donut charts, radar charts, histograms.
AI Maritime Copilot	Implemented	<i>CODE_PH_0, CODE_PH_1</i>	LangGraph 3-tier cognitive query dispatcher with 5 specialized agent roles.
Executive Command Center	Implemented	<i>CODE_PH_0, CODE_PH_1</i>	Real-time KPIs, berth utilization, live AIS vessel map, 30-day revenue/throughput trend.
Monte Carlo Policy Simulator	Implemented	<i>CODE_PH_0, CODE_PH_1</i>	1,000 statistical iterations calculating P10, P50, and P90 revenue quantiles.
Digital Twin Scenario Modeler	Implemented	<i>CODE_PH_0, CODE_PH_1</i>	Virtual port stress-testing for cargo surges, weather cyclones, and delays.
User Authentication / RBAC	Not Implemented	N/A	Intentionally omitted in current PoC build; all endpoints are publicly accessible.
SQL Relational Database Storage	Partially Implemented	N/A	Platform uses Pandas In-Memory Data Pool backed by CSV persistent files.